

Windows Functions Assignment

Assumed Table

We'll assume your Superstore table, Please download the superstore table from google.
Answer Below questions using superstore table.

A. Ranking Functions (ROW_NUMBER, RANK, DENSE_RANK)

1. Find the top 3 highest sales orders per region using ROW_NUMBER().
2. Rank customers within each region based on their total sales using RANK().
3. Find customers who have the same rank (based on sales) in each region using DENSE_RANK().
4. Get the 5th highest order (by Sales) in each Category using ROW_NUMBER().
5. Find the top 10 most profitable products overall using any ranking window function.

B. Aggregate Window Functions (SUM, AVG, COUNT, etc.)

6. Calculate running total of sales for each region ordered by order date.
7. Find cumulative profit for each customer based on order date using SUM() OVER (PARTITION BY ...).
8. Show average sales per category along with each order using AVG() OVER.
9. Show each order's sales and the total category sales in the same row using SUM() OVER (PARTITION BY Category).
10. Calculate percentage contribution of each sub-category to its category's total sales.

C. Analytical Window Functions (LAG, LEAD, FIRST_VALUE, LAST_VALUE)

11. Find the previous order date for each customer using LAG(OrderDate).
12. Find the next product purchased by each customer using LEAD(ProductName).
13. Calculate change in profit compared to previous order for each customer.
14. Display the first product ordered by each customer using FIRST_VALUE().
15. Show the latest (most recent) order amount per customer using LAST_VALUE().

D. NTILE and Advanced Ranking

16. Divide all customers into 4 quartiles based on their total sales using NTILE(4).
17. Identify the top 25% of orders by profit using NTILE(4) and filtering the top quartile.
18. Find the median sales per category using a combination of NTILE(2) or PERCENT_RANK().

E. Mixed & Analytical Use Cases

19. For each region, find customers whose sales are above the regional average using `AVG()` `OVER`.
20. For each category, find the difference between each order's sales and the category's average sales.